Wazuh Home Lab

Phase 4: pfSense Virtual Firewall Configuration

Introduction:

This phase extends the Wazuh SOC home lab by introducing a pfSense virtual firewall as the network gateway for all virtual machines. Building on the host-based monitoring and threat intelligence capabilities established in previous phases, this stage adds a dedicated firewall layer to control and monitor inter-VM and internet-bound traffic.

pfSense was deployed as a VirtualBox VM with dual network interfaces — a WAN interface for internet access via NAT and a LAN interface for internal VM connectivity via a Host-Only adapter. This configuration simulates a real-world network perimeter where all traffic flows through a central firewall before reaching internal hosts or the internet.

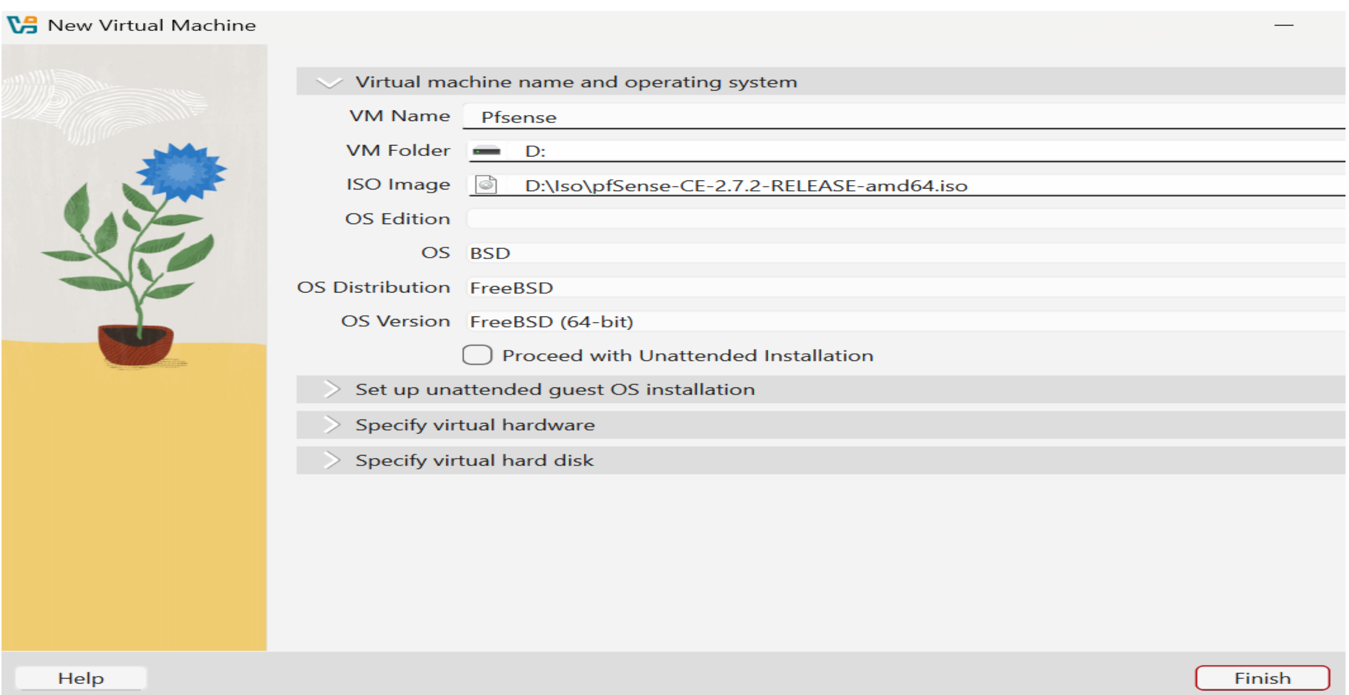
Objective:

The objective of this phase was to configure pfSense as the default gateway for the lab's virtual machines and integrate it with the existing Wazuh deployment for centralized firewall alerting.

- Deploy pfSense as a VirtualBox virtual machine
- Configure WAN and LAN network interfaces
- Route all VM traffic through pfSense as the default gateway
- Create and test a firewall rule to block ICMP (ping) from VMs to the internet
- Integrate pfSense with Wazuh for centralized alert monitoring

1) pfSense Installation:

pfSense was obtained as an ISO image and installed as a new VirtualBox VM. During the initial installation attempt, a boot loop was encountered where the installer repeatedly restarted without completing the setup.

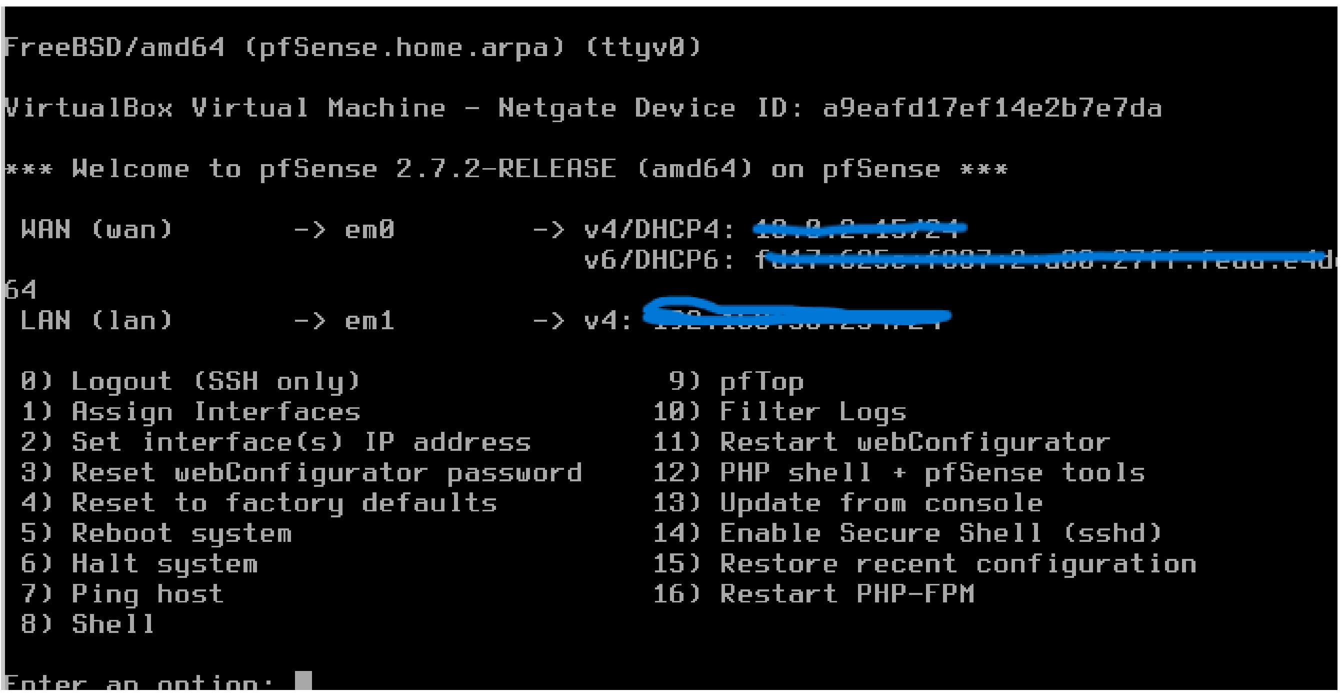


1.1) Installation Boot Loop Issue:

The installation entered a continuous loop, preventing the setup from completing. This is a known VirtualBox/pfSense behavior where the installer attempts to reboot from the ISO after the initial stage rather than proceeding from the installed disk.

1.2) Resolution:

- Monitored the installation progress in the VM console
- Detached the ISO image from VirtualBox storage settings mid-installation, immediately after the installer finished writing to disk
- VM rebooted from the installed disk and completed configuration successfully



2) Network Configuration:

pfSense was configured with two virtual network interfaces to separate WAN (internet-facing) and LAN (internal) traffic.

The WAN interface was configured using VirtualBox's NAT adapter, providing the pfSense VM with outbound internet access through the host machine. This interface was automatically assigned the IP address via DHCP from VirtualBox's internal NAT service.

The LAN interface was configured on a VirtualBox Host-Only network. pfSense was assigned the static IP which serves as the default gateway for all virtual machines on this subnet. DHCP was disabled and all VMs were configured with static IP addresses manually.

3) Virtual Machine Gateway Configuration:

Each virtual machine was configured to use the pfSense LAN interface as its default gateway, ensuring all outbound traffic routes through the firewall.

3.1) Linux VMs (Kali Linux / Linux Mint):

The default gateway was configured via the command line by removing the existing default route and replacing it with the pfSense LAN address.

3.2) Windows 10 VM:

Windows 10 was configured with a static IPv4 address and the pfSense LAN IP was set as the default gateway through the Network Adapter settings (Control Panel → Network and Sharing Center → IPv4 Properties).

4) Firewall Rule: Block ICMP to Internet:

4.1) Rule Configuration:

A firewall rule was created on the pfSense LAN interface to block outbound ICMP (ping) traffic from virtual machines destined for the internet. This was configured through the pfSense web interface under Firewall → Rules → LAN.

- Action: **Block**
- Interface: **LAN**
- Protocol: **ICMP**
- Source: **LAN subnet**
- Destination: **Any (internet)**
-

☐	✕	0/0 B	IPv4 ICMP any	*	*	*	*	*	none	ping block

4.2) Testing:

After applying the rule, ping tests were conducted from VMs to external addresses (8.8.8.8 and google.com). All ICMP requests were successfully blocked by pfSense, confirming the rule was functioning as intended. Pings to the pfSense LAN interface remained operational, confirming only internet-bound ICMP was affected.

```
wazuhserver@wazuhserver-VirtualBox:~$ ping  -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
```

5) Wazuh Integration:

pfSense was integrated with the Wazuh Manager to forward firewall logs and generate security alerts on the Wazuh dashboard, extending the lab's centralized monitoring capability to include network-level firewall events.

5.1) Implementation:

- Configured pfSense to forward syslog events to the Wazuh Manager

```
wazuhserver@wazuhserver-VirtualBox: ~
GNU nano 7.2                    /var/ossec/etc/ossec.conf *
<!-- Choose between "plain", "json", or "plain,json" for the format of intern>
<logging>
  <log_format>plain</log_format>
</logging>

<remote>
  <connection>secure</connection>
  <port>1514</port>
  <protocol>tcp</protocol>
  <queue_size>131072</queue_size>
</remote>

<remote>
  <connection>syslog</connection>
  <port>514</port>
  <protocol>udp</protocol>
  <allowed-ips>1[redacted]</allowed-ips>
</remote>

^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute     ^C Location
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line
```

- Wazuh Manager was configured to receive and parse pfSense log format
- Firewall alerts became visible on the Wazuh dashboard

```
<!-- Rule for PFsense -->
<rule id="100005" level="5">
 <match>filterlog</match>
 <same_source_ip/>
 <description>pfSense firewall event</description>
</rule>
```

5.2) Result:

After integration, pfSense firewall events — including blocked connections and interface activity — appeared as alerts in the Wazuh dashboard. This provides SOC-level visibility into network boundary events alongside the existing host-based monitoring from previous phases.

500 Matched Firewall Log Entries. (Maximum 500)

Action	Time	Interface	Rule	Source
✗	Apr 19 21:41:15	LAN	Default deny rule IPv6 (1000000105)	[fe80:
✗	Apr 19 21:41:15	LAN	Default deny rule IPv6 (1000000105)	[fe80:
✗	Apr 19 21:41:15	LAN	Default deny rule IPv6 (1000000105)	[fe80:
✗	Apr 19 21:41:15	LAN	Default deny rule IPv6 (1000000105)	[fe80:
✗	Apr 19 21:41:15	LAN	Default deny rule IPv6 (1000000105)	[fe80:
✗	Apr 19 21:41:15	LAN	Default deny rule IPv6	[fe80:

Conclusion:

In this phase, the SOC home lab was significantly enhanced by deploying pfSense as a virtual firewall gateway. pfSense was configured with WAN and LAN interfaces, routing traffic for Windows 10, Kali Linux, and Linux Mint VMs through a controlled network boundary. A firewall rule was created and verified to block internet-bound ICMP traffic, demonstrating practical firewall policy enforcement.

Integration with Wazuh Manager extended the lab's monitoring coverage to include firewall events, further simulating a real-world SOC environment. Combined with the host-based monitoring, network intrusion detection, and threat intelligence enrichment from previous phases, this phase completes a multi-layered security monitoring stack encompassing endpoint, network, and perimeter visibility.